From Reactive Administration to Predictive Intelligence

The Legacy Constraints

Manual Selection
Administrators manually guess caregiver matches based on limited data, leading to poor compatibility.

Late Risk Discovery
Issues like isolation, neglect, or health decline are only detected after they escalate.

Static Service
A one-size-fits-all approach where every elder receives identical care parameters regardless of individual behavior.

Care Pal is not just a CRUD management system. It is an AI-First Decision Engine.

AI

The AI Evolution

Intelligent Matching
Algorithms analyze skills, location, health history, and personality to generate compatibility scores.

Preventative Alerts
The system monitors communication patterns and payment behaviors to predict risks before they become emergencies.

Dynamic Personalization
Care plans automatically adjust based on usage patterns and behavioral feedback.

Transforming elderly care in the MENA region by replacing manual guesswork with data-driven safety.

Seven Pillars of the Artificial Intelligence Engine

Intelligent Matching Logic

Replaces manual assignment by scoring compatibility between Elder profiles (needs, personality) and Caregiver data (skills, experience).

Early Risk Prediction

Analyzes service history, missed visits, and payment anomalies to forecast isolation, neglect, or service failure risks.

Natural Language Processing (NLP)

Scans chat logs and textual feedback to detect emotional states (sadness, stress, anger), triggering alerts for negative sentiment trends.

Automated Emergency Response

Classifies emergency button triggers and automatically routes requests to police or ambulance simulations while marking the case as high priority.

Smart Admin Insights

Generates automated quality reports and operational alerts, reducing the manual monitoring workload for platform administrators.

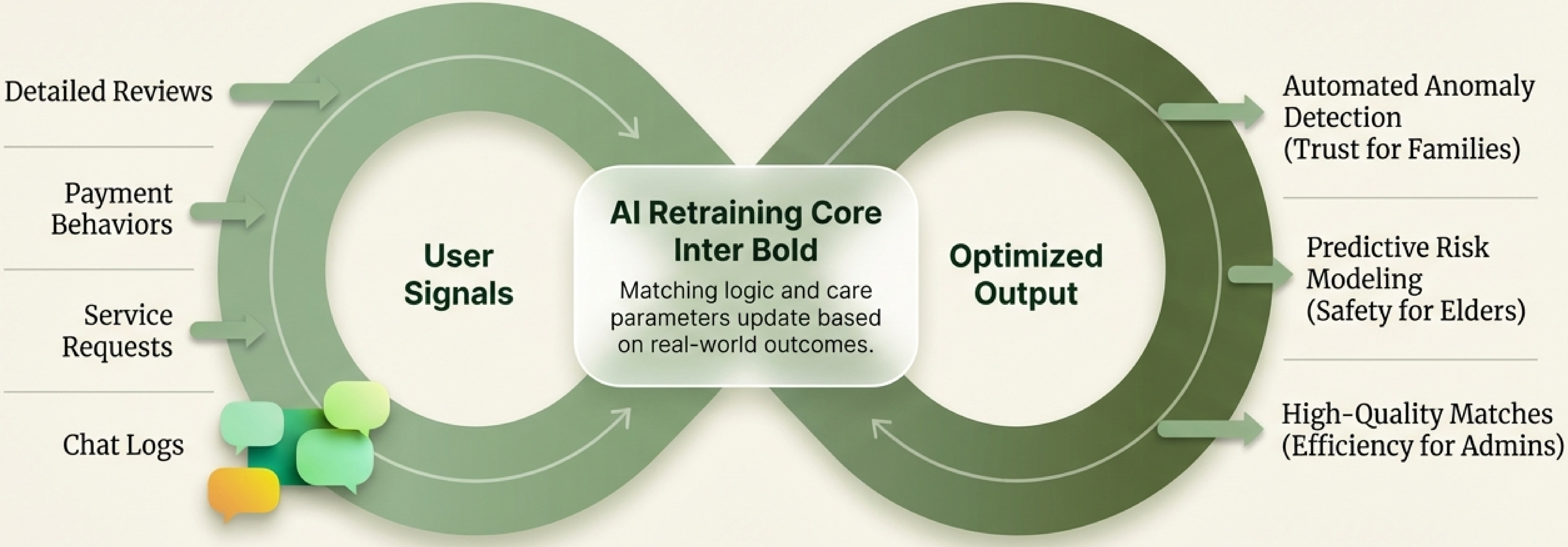
Personalized Care Plans

Dynamically builds and adjusts care profiles based on individual behavior, visit frequency, and service consumption.

Adaptive Learning

Utilizes ratings, reviews, and payment data to continuously retrain the model, improving future matching accuracy and service quality.

A Self-Learning Ecosystem for Safer Care



Every interaction fuels the engine, ensuring Care Pal evolves from a service provider into a proactive guardian.

Focus: Non-technical users in Egypt/MENA. Arabic & English Support. **Boundaries:** Service intelligence only; no medical diagnosis or real-time IoT.

The Tech Stack: How AI Powers Each Feature

Intelligent Matching (RAG Architecture)

The Stack
Llama-3-8b (Groq) + all-MiniLM-L6-v2 + Pinecone

The Logic
Retrieval: Searches Vector DB for skill matches.
Generation: LLM analyzes psychological fit (e.g., patience for Alzheimer's) to assign a Compatibility Score.

Automated Emergency (Autonomous Agents)

The Stack
LangChain (Agents) + Llama-3-70b

The Logic
Classification: Detects intent (Medical vs. False Alarm).
Action: Triggers Decision Tool to execute API calls (Twilio) for simulated dispatch.

Personalized Care (Context-Aware)

Model: GPT-4o or Mixtral-8x7b

Ingests unstructured medical reports to generate dynamic daily schedules (medication, mental games) rather than generic templates.

Emotion & NLP (Sentiment Engine)

Model: DistilBERT or Llama-3

Real-time scanning of chat logs assigns Sentiment Scores (-1 to +1). Drops in score trigger "Psychological Risk Alerts" to the dashboard.

Self-Learning (Dynamic RAG)

Tech: Feedback Vectorization

Completed service feedback is vectorized and pushed to Pinecone, updating system "memory" instantly without full model retraining.